The Quantum Action

Relation to Feynman Diagrams

we start this discussion by defining a quantum action:

$$\Gamma(\varphi) \equiv \frac{-1}{2}\int \frac{d^dk}{(2\pi)^d}\,\tilde{\varphi}(-k)(k^2+m^2-\Pi(k^2))\,\tilde{\varphi}(k) + \sum_{n=3}^{\infty}\frac{1}{n!}\int\frac{d^dk_1}{(2\pi)^d}\cdots\frac{d^dk_2}{(2\pi)^d}(2\pi)^d\delta^d(k_1+\dots+k_n)\,V_n(k_1,\dots k_n)\,\tilde{\varphi}(k_1)\dots\tilde{\varphi}(k_n)$$

Here, $\tilde{\varphi}(k) = \int d^dx\, e^{-ikx}\varphi(x)$

This action is also called the effective or quantum effective action.

This action is defined such that the tree-level Feynman diagrams it generates give the complete scattering amplitude of the original theory.

The point of this discussion is to relate $\Gamma(\varphi)$ to $W(J)$ where:

$$Z(J) = \exp[iW(J)]\ ;\ Z(J) = \int D\varphi\,\exp[iS(\varphi) + i\int d^dx\, J\varphi]\ ;\ S = \int d^dx\,\mathcal{L}$$

Q: J is termed the classical source implying it creates a particle but I thought the premise of QFT was that particles are created by creation and annihilation operators. How does this work?

Now consider the path integral corresponding to the Quantum Action:

$$Z_\Gamma(J) \equiv \int D\varphi\,\exp[i\Gamma(\varphi) + i\int d^dx\, J\varphi] = \exp[iW_\Gamma(J)]$$

Sum of connected diagrams w. c. props and c. vertices.

The tree level contribution of $W_\Gamma(J)$ will be equal to $W(J)$.

So how do we isolate the tree-level contributions?? This can be done with a trick, lets start by introducing the dimensionless parameter $\hbar$:

$$Z_{\Gamma,\hbar}(J) \equiv \int D\varphi\,\exp\left[\frac{i}{\hbar}\left(\Gamma(\varphi) + \int d^dx\, J\varphi\right)\right] = \exp[iW_{\Gamma,\hbar}(J)]$$

When we draw the diagrams for this Path Integral, we find that every propagator has a factor of $\hbar$, every source and vertex has $1/\hbar$.

How did we get this?

$$Z_{\Gamma,\hbar}(J) = \int D\varphi\,\exp\left[\frac{i}{\hbar}\left(\Gamma(\varphi) + \int d^dx\, J\varphi\right)\right]$$

$$= \int D\varphi\,\exp\left[\frac{i}{\hbar}\left(\frac{-1}{2}\frac{d^dk}{(2\pi)^d}\,\tilde{\varphi}(-k)(k^2+m^2-\Pi(k^2))\,\tilde{\varphi}(k) + \underbrace{\sum_{n=3}^{\infty}\frac{1}{n!}\int\frac{d^dk_1}{(2\pi)^d}\cdots\frac{d^dk_n}{(2\pi)^d}(2\pi)^d\delta^d(k_1+\dots+k_n)\,V_n(k_1,\dots k_n)\,\tilde{\varphi}(k_1)\dots\tilde{\varphi}(k_n)}_{\tilde{\mathcal{L}}_{int}} + \int d^dx\, J\varphi\right)\right]$$

$$= \exp\left[\frac{i}{\hbar}\tilde{\mathcal{L}}_{int}\left(\frac{\delta}{\delta\tilde{J}(k)}\right)\right]\int D\varphi\,\exp\left[\frac{i}{\hbar}\left(\frac{-1}{2}\frac{d^dk}{(2\pi)^d}\,\tilde{\varphi}(-k)(k^2+m^2-\Pi(k)^2)\,\tilde{\varphi}(k) + \int d^dx\, J\varphi\right)\right]$$

Roughly of this form

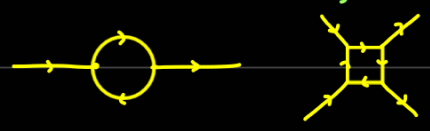
$$= \exp\left[\frac{i}{\hbar}\tilde{S}_{int}\left(\frac{\delta}{\delta \tilde{J}(k)}\right)\right]\cdot\exp\left[\frac{i}{\hbar}\int d^4x\, d^4x'\, J(x)\, \Delta(x-x')\, J(x')\right]$$

This gives us the overall $\hbar$ factor: $\hbar^{P-E-V}$.

The number of closed loops will be given by: $P-E-V = L-1$

This can be thought of as: Assign an unfixed momentum to each internal line. There are $P-E$ internal lines.

Now, lets find out how many of these momenta are fixed.



Vertices provide V constraints. However a linear combination of these vertices only ensures momentum conservation. Thus we must subtract one constraint.

This gives us $P-E-V+1 = L$

We can expand $\omega_{\Gamma,\hbar}(J)$ in a power series in $\hbar$ to get:

$$\omega_{\Gamma,\hbar}(J) = \sum_{L=0}^{\infty}\hbar^{L-1}\omega_{\Gamma,L}(J)$$

As we take the limit $\hbar\to 0$, we get:

$\omega(J) = \omega_{\Gamma,L=0}(J)$ $\longrightarrow$ In the limit $\hbar\to 0$, $L=0$ is the dominant term.

Q: Why are we taking $\hbar$ to 0? Doesn't this basically make the path integral 1?

We now want to equate $Z_{\Gamma,\hbar}$, the best way to do so is by the method of stationary phase.

The stationary point (field configuration) is given by the solution to the quantum equation of motion.

$$\frac{\delta}{\delta\varphi(x)}\Gamma(\varphi) = \partial_\mu\frac{\delta}{\delta\partial_\mu\varphi}J(x)\varphi(x)\cdot\frac{\delta}{\delta\varphi(x)} - J(x)\varphi(x) = -J(x)$$

$\longrightarrow$ We can do this because all the φ's in Γ are Fourier transformed :)

If we let $\varphi_J(x)$ denote the solution of the above equation, the stationary phase approximation becomes:

$$Z_{\Gamma,\hbar}(J) = \exp\left[\frac{i}{\hbar}\left(\Gamma(\varphi_J) + \int d^dx\, J\varphi_J\right) + O(\hbar^0)\right]$$

Order $\hbar^{-1}$ $\longrightarrow$ $L=0$

With all the above results, we can conclude that:

$$\omega(J) = \Gamma(\varphi_J) + \int d^dx\, J\varphi_J$$

Further Analysis

We know that the vacuum expectation of the field operator is given by:

$$\langle 0|\varphi(x)|0\rangle = \frac{\delta}{\delta J(x)}\omega(J)\Big|_{J=0}$$

Now, we may ask the question, What if we do not set $J=0$?

$$\langle 0|\varphi(x)|0\rangle_J = \frac{\delta}{\delta J(x)} W(J) = \frac{\delta}{\delta J(x)}\left[\Gamma(\varphi_J) + \int d^dx\, J\varphi_J\right] = \frac{\delta}{\delta J(x)}\Gamma(\varphi_J) + \varphi_J(x) + \int d^dy\, J(y)\frac{\delta\varphi_J(y)}{\delta J(x)}$$

Using the chain rule on the quantum action gives us:

$$\frac{\delta}{\delta J(x)}\Gamma(\varphi_J) = \int d^dy\, \frac{\delta\Gamma(\varphi_J)}{\delta\varphi_J(y)}\frac{\delta\varphi_J(y)}{\delta J(x)}$$

Plugging it in:

$$\langle 0|\varphi(x)|0\rangle_J = \int d^dy\left[\frac{\delta\Gamma(\varphi_J)}{\delta\varphi_J(y)} + J(y)\right]\frac{\delta\varphi_J(y)}{\delta J(x)} + \varphi_J(x) = \int d^dy\left[-J(y) + J(y)\right]\frac{\delta\varphi_J(y)}{\delta J(x)} + \varphi_J(x) = \varphi_J(x)$$

Thus the vacuum expectation value of the field operator in presence of a non-zero source is equal to the solution of the quantum equation of motion.

It is also possible to write the quantum action in terms of a derivative expansion:

Ordinary Functions

Infinite no. of terms with more and more derivatives.

$$\Gamma(\varphi) = \int d^dx\left[-\mathcal{U}(\varphi) - \frac{1}{2}Z(\varphi)\partial^\mu\varphi\,\partial_\mu\varphi + \dots\right]$$

Quantum Potential / Effective Potential / Quantum effective potential.